

Question 1

SIMULTANEOUS

EQUATIONS

a) $x + y = 10$ — (1)
 $3x - 4y = 2$ — (2)

$3 \times (1) : 3x + 3y = 30$ — (3)

(3) - (2)

$3x + 3y - (3x - 4y) = 30 - 2$

$3x + 3y - 3x + 4y = 28$

$7y = 28$

$y = 4$

sub into (1)

$x = 6$

b) $x + 5y = 15$ — (1)
 $-x + 2y = 6$ — (2)

(1) + (2)

$7y = 21$

$y = 3$

$x = 0$

c) $2x + y = -8$ — (1)
 $3x - 2y = -12$ — (2)

$2 \times (1) : 4x + 2y = -16$ — (3)

(3) + (2)

$7x = -28$

$x = -4$

$y = 0$

d) $x + 3y = 6$ — (1)
 $2x + 4y = 8$ — (2)

rearrange (1)

$x = 6 - 3y$ — (3)

sub (3) into (2)

$2(6 - 3y) + 4y = 8$

$12 - 6y + 4y = 8$

$-2y = -4$

$y = 2$

$x = 0$

e) $2x - 5y = 11$ — (1)
 $5x - 2y = 17$ — (2)

$5 \times (1) : 10x - 25y = 55$

$2 \times (2) : - (10x - 4y = 34)$

$-21y = 21$

$y = -1$

sub into (1)

$x = 3$

f) $3x + y = 7$ — (1)
 $x + 2y = 9$ — (2)

rearrange (2) $x = 9 - 2y$ — (3)

sub (3) into (1)

$3(9 - 2y) + y = 7$

$27 - 6y + y = 7$

$-5y = -20$

$y = 4$

sub into (1)

$x = 1$

Question 2

a) $3x + 5y = 25$ — ①
 $2x - y = 8$ — ②

$5 \times ②$ $10x - 5y = 40$ — ③

① + ③ $13x = 65$

$x = 5$

sub into ② $y = 2$

b) $4x + 7y = 19$ — ①
 $3x - y = 8$ — ②

rearrange ②

$y = 3x - 8$ — ③

sub ③ into ①

$4x + 7(3x - 8) = 19$

$4x + 21x - 56 = 19$

$25x = 75$

$x = 3$

sub into ②

$y = 1$

c) $2x - 5y = 13$ — ①
 $5x - 3y = -15$ — ②

$5 \times ①$ $10x - 25y = 65$

$2 \times ②$ $10x - 6y = -30$

① - ② $-19y = 95$

$y = -5$

sub into ①

$x = -6$

d) $4x - 3y = 9$ — ①
 $-x + 3y = 6$ — ②

① + ② $3x = 15$
 $x = 5$

sub into ①

$y = \frac{11}{3}$